

How to Run NVIDIA Nemotron 3 Ultra for FREE in Your Terminal

This guide shows you how to use NVIDIA's Nemotron 3 Ultra AI model with OpenCode, a terminal-based coding agent. You do not need a powerful GPU because the model runs on NVIDIA's cloud servers.

What You Need

- A computer with a terminal (Command Prompt, PowerShell, or Terminal)
 - A free NVIDIA account
 - OpenCode installed
 - An NVIDIA API key
 - Internet connection
 - No local GPU required
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Step 1: Get Your NVIDIA API Key

1. Go to the NVIDIA Nemotron 3 Ultra model page.
 2. Sign in or create a free NVIDIA account.
 3. On the model page, click Generate API Key.
 4. Copy the API key. It will look like: nvapi-xxxxxxxxxxxxxxxxxx
 5. Never share this key with anyone.
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Step 2: Install OpenCode

Open your terminal and run one of these commands:

For Mac/Linux/Git Bash:

```
bash
```

```
curl -fsSL https://opencode.ai/install | bash
```

For Windows (using npm):

```
bash
```

```
npm install -g opencode-ai
```

If you don't have Node.js, install it first from nodejs.org.

Step 3: Connect NVIDIA to OpenCode

1. Open your terminal and go to your project folder:

2. `bash`
 3. `cd /path/to/your/project`
 4. Start OpenCode:
 5. `bash`
 6. `opencode`
 7. Inside OpenCode, type:
 8. `text`
 9. `/connect`
 10. Select NVIDIA from the list.
 11. Paste your NVIDIA API key when asked.
- OpenCode will save the key securely.
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Step 4: Select Nemotron 3 Ultra Model

Inside OpenCode, type:

`text`

`/models`

Then search and select:

`text`

`nvidia/nemotron-3-ultra-550b-a55b`

This is the official model ID for Nemotron 3 Ultra.

Step 5: Start Building

Now you can give OpenCode coding tasks. For example:

Build a premium landing page with a glossy 3D object. Use a modern, polished UI with smooth animations, responsive design, and production-quality code.

OpenCode will use Nemotron 3 Ultra to plan, write code, and even create files for you.

Optional: Manual Configuration

Instead of using `/connect`, you can manually edit the configuration file at:

`text`

`~/.config/opencode/opencode.json`

Add this JSON:

`json`

`{`

`"$schema": "https://opencode.ai/config.json",`

`"provider": {`

`"nvidia": {`

`"npm": "@ai-sdk/openai-compatible",`

`"name": "NVIDIA NIM",`

`"options": {`

`"baseUrl": "https://integrate.api.nvidia.com/v1",`

```
    "apiKey": "nvapi-YOUR_KEY_HERE"
  },
  "models": {
    "nvidia/nemotron-3-ultra-550b-a55b": {
      "name": "Nemotron 3 Ultra",
      "limit": {
        "context": 1000000,
        "output": 16384
      }
    }
  }
}
```

Replace nvapi-YOUR_KEY_HERE with your actual API key. Then launch OpenCode and use /models to select the model.

Important Details

- Model ID: nvidia/nemotron-3-ultra-550b-a55b
- NVIDIA API endpoint: <https://integrate.api.nvidia.com/v1>
- Context window: Up to 1,000,000 tokens (very large memory)
- Total parameters: 550B–561B (billion)
- Active parameters: 55B (makes it fast)
- Best for: Coding, planning, reasoning, tool calling, long-context tasks

Note: The free endpoint has limits (about 40 requests per minute). NVIDIA may change or stop the free tier at any time.

Security Tips

- Keep your API key secret. Do not share it or upload it to GitHub.
 - Do not send personal or sensitive data to the AI.
 - OpenCode runs a local server in the background. Close it when not in use.
 - Always use the latest version of OpenCode: `npm update -g opencode-ai`
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Troubleshooting

- If opencode command is not found, run `npm install -g opencode-ai` again.
 - If the model does not load, check your opencode.json file for typos.
 - If you get rate limit errors, wait a minute and try again.
 - For more help, visit the official OpenCode documentation.
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Official Links

- NVIDIA Nemotron 3 Ultra: [NVIDIA Model Page](#)
- OpenCode Documentation: <https://opencode.ai/docs>
- OpenCode NVIDIA Provider Guide: [OpenCode Docs](#)

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